

Quantum Recognition Heuristics: A Tutorial Example of Quantum Heuristics

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Abstract—We present a tutorial example of the Quantum Heuristics Toolbox, a new research direction of quantum cognition. Quantum cognition is an emerging field that applies the matrix formalism of quantum theory to model context-sensitive aspects of human reasoning, while remaining agnostic about whether the brain functions as a quantum computer. The toolbox serves two core purposes. First, it offers a descriptive account of cognitive biases in human heuristic or rule-based decision making that are difficult to reconcile with classical decision theory without introducing strong auxiliary assumptions. Second, it provides a foundation for developing bounded-rational, human-like artificial intelligence by encoding heuristics as quantum algorithms. This poster focuses on a quantum extension of the recognition heuristic—a simple yet powerful rule: when one of two alternatives is recognized and the other is not, people tend to choose the recognized option, regardless of the context. We demonstrate how this heuristic can be implemented probabilistically as a quantum-controlled unitary operator. Crucially, the quantum model captures hindsight bias through projective measurement effects, an effect that challenges Bayesian belief update model without additional ad-hoc assumptions about memory and context. We conclude by outlining how this framework can be extended to model interference, order, and supercorrelation effects in multi-cue heuristics reasoning.

Index Terms—Quantum Cognition; Decision Making; Quantum Logic; Quantum Application

I. INTRODUCTION

In this poster, we introduce a new direction in quantum cognition: the quantum heuristics toolbox. Quantum cognition is an emerging field that uses the mathematical formalism of quantum theory—not to suggest that the brain is a quantum computer, but as a descriptive framework for modeling context-sensitive patterns in human reasoning. This approach accounts for various cognitive phenomena such as order effects, interference effects, and violations of the classical law of total probability—patterns that traditional probabilistic models struggle to explain [1]–[3].

The quantum heuristics toolbox aims to unify two foundational modes of reasoning: heuristic reasoning, which relies on simple, rule-based strategies, and probabilistic reasoning, which addresses uncertainty and variability. For decades, research has shown that people often rely on fast and frugal heuristics, especially under constraints of time or resources [4], [5]. An example is the recognition heuristic, where individuals favor recognized options over unrecognized ones independent

of question asked [6]. These strategies, though simple, are often highly effective in real-world settings.

However, traditional models treat heuristics as deterministic processes and often fail to capture the probabilistic variation seen in actual decision-making. Efforts to reconcile heuristics and probabilistic reasoning—such as incorporating heuristics into sequential sampling models [7] or adding noise to cue processing [8], [9]—typically treat them as distinct components. These models do not fully explain how people fluidly transition between rule-based shortcuts and uncertain inference.

The quantum heuristics toolbox offers a unified solution by modeling heuristics as quantum-controlled unitary operations—the quantum counterpart of classical “if-then” rules [10], [11]. When a heuristic cue (the control qubit) is measured, the outcome is deterministic, mirroring classical heuristic behavior. But when unmeasured, the cue remains in a superposition, and sampling algorithms such as decision diffusion models (which are the most successful models in explaining human stochastic behaviors) are needed to estimate such superpositions [12], [13].

Finally, beyond cognitive modeling, this approach opens avenues for building bounded-rational quantum AI. Human heuristics reflect bounded rationality—making good decisions with limited resources. Quantum computing is often motivated by the promise of improved efficiency over classical computation. Thus, marrying the two fulfills the goal of each others. Notably, many aspects of heuristic reasoning resemble the matrix-based probabilistic calculus found in quantum theory, even if the underlying cognitive mechanisms are classical. This structural parallel suggests the toolbox may not only advances our understanding of human cognition but also lays groundwork for developing more adaptive, human-like AI systems.

II. QUANTUM RECOGNITION HEURISTICS

As a simple tutorial, this poster focuses on a quantum extension of the recognition heuristic, defined by the following equation assuming the target state is an equal superposition ψ_0 :

$$\begin{aligned} RH = & M_0 \otimes (M_1 \otimes (X\bar{H}) + M_0 \otimes R(\theta)) \\ & + M_1 \otimes (M_0 \otimes \bar{H} + M_1 \otimes I_2), \end{aligned} \quad (1)$$

where $M_0 = \text{diag}([1 \ 0])$ and $M_1 = \text{diag}([0 \ 1])$ are projectors onto $|0\rangle$ and $|1\rangle$, $X\bar{H}$ is the Hadamard Flip gate, $\bar{H}$ is the Hadamard gate, $R(\theta)$ is a 2D rotation matrix, and I_2 is the identity. The RH gate is controlled by two qubits: if both are $|1\rangle$, it applies a rotation to ψ_0 ; if the first is $|1\rangle$ and the second $|0\rangle$, it maps ψ_0 to $|0\rangle$ via $X\bar{H}$; if the first is $|0\rangle$ and the second $|1\rangle$, it maps ψ_0 to $|1\rangle$ via $\bar{H}$; and if both are $|0\rangle$, it leaves ψ_0 unchanged. These operations reproduce the outcomes of the classical deterministic recognition heuristic. When the control qubits are in superposition, however, probabilistic behavior emerges. A circuit diagram and a corresponding “truth table” are provided in the “Quantum Recognition Heuristics” section of the poster.

III. EMPIRICAL TEST OF QUANTUM HEURISTICS

Quantum heuristics extend deterministic fast and frugal heuristics in two key ways. First, they incorporate probabilistic dynamics, allowing the model to account for variability in human responses. Second, when integrated with sampling algorithms such as the quantum sequential sampler, they can predict graded probability judgments (e.g., how likely one city is larger than another), which is ignored in traditional heuristic research. These two advantages, however, can also be similarly captured by a classical probabilistic extension of the heuristics. So where do the ‘quantum-likeness’ in human heuristics reasoning come from?

For single-cue heuristics like the recognition heuristic, one example of “quantum-likeness” is hindsight bias observed in the ABA experimental paradigm where a question is asked twice with an intervening question in between, conceptually analogous to a Stern-Gerlach experiment [14]. The “Why Quantum” section of the poster illustrates this setup. Hindsight bias arises when participants revise their confidence or probability judgment about a prior answer after encountering the intermediate question. In the poster’s example, the probability of recognition of “Frankfurt” increases significantly after answering “Which city has a larger population, Frankfurt or Mainz?” compared to the initial judgment made beforehand.

Such hindsight bias naturally arises from projective measurements after responding the intermediate question in the quantum recognition heuristics. The poster illustrates this with a concrete calculation example in the “Why Quantum” section. Classical or Bayesian models cannot easily account for this effect, as answering B introduces no new information that would update beliefs about A. While a Bayesian model incorporating context or working memory might explain the bias, it would require additional, potentially untestable assumptions beyond standard Bayesian inference.

In multiple-cue processing, three additional quantum-like effects may emerge in heuristic reasoning. First, interference effects can occur—for example, when: $P(\text{choose A \& Cue 1 presented \& Cue 2 presented})$ is greater than $P(\text{choose A \& Cue 2 presented})$ [13]. Such a violation is impossible under classical set-theoretic probability but permitted in quantum theory when the events are incompatible and participants process the more likely event first [15]. Second, order effects

may arise, where the sequence of cue presentation alters the probability distribution leading to $P(\text{choose A} \mid \text{Cue 1 presented first \& Cue 2})$ not equivalent to that given Cue 2 presented first & Cue 1 [16]. Third, super-correlations may appear when cues that should be independent violate the CHSH inequality—suggesting hidden correlations. While not proof of entanglement or non-locality, such correlations can be effectively modeled as entangled states in quantum computation [17].

In summary, while these effects do not imply that the human mind operates using quantum logic, they challenge classical models and support the value of a non-classical, quantum-inspired account of cognition. Again, their normative basis may not stem from physical quantumness in the brain, but could reflect bounded-rational reasoning. Still, the structural parallels between human behavior and quantum logic offer a promising foundation for developing human-like quantum algorithms in future artificial intelligence.

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